

Introduction to Message Passing Interface (MPI)

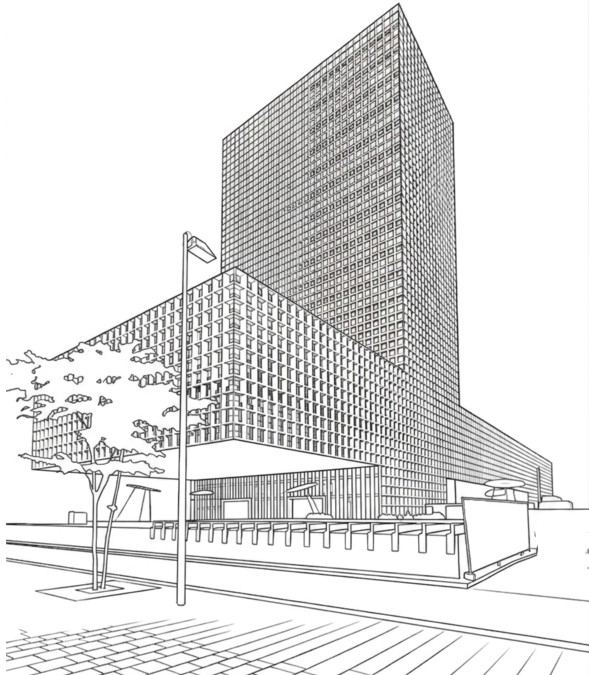
Lesson 2

Matteo Barborini

University of Luxembourg - HPC Platform

matteo.barborini@uni.lu

23rd of September 2026



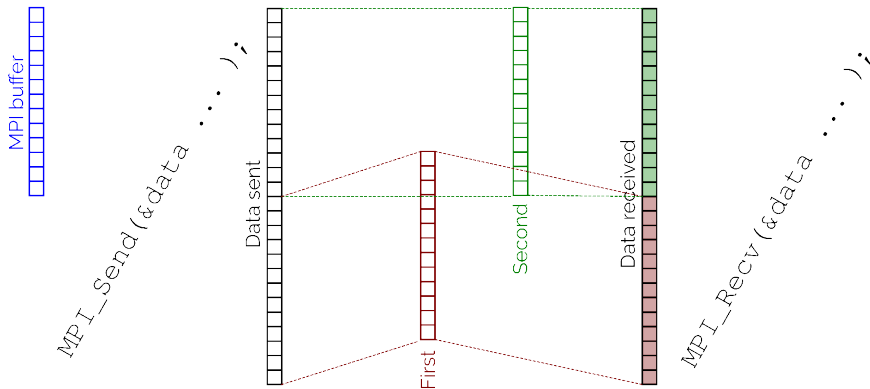
MPI Send and MPI Recv Synchronizations

```
1 int MPI_Send(const void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int destination_rank,  
5             int tag,  
6             MPI_Comm comm);
```

```
1 int MPI_Recv(void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int source_rank,  
5             int tag,  
6             MPI_Comm comm,  
7             MPI_Status *status);
```

MPI_Send is locally blocking, meaning that the routine MIGHT NOT RETURN until the memory holding the message is available to reuse. This depends on the difference between the information that must be sent and the buffer of the communications.

MPI_Recv is blocking, meaning that the routine RETURNS only after the receive buffer contains the newly received message.



```

1 int MPI_Bsend(const void *data,
2             int data_size,
3             MPI_Datatype data_type,
4             int destination_rank,
5             int tag,
6             MPI_Comm comm);

```

```

1 // Legacy Signature
2 int MPI_Buffer_attach(void *buffer,
3                       int size);
4
5 // MPI_Count (MPI-4) Is used for
   datacounts exceeding 32-bit integer
   limits
6 int MPI_Buffer_attach_c(void *buffer,
7                       MPI_Count size);
8
9 // Safely return the buffer memory
10 int MPI_Buffer_detach(void *buffer,
11                      int *size)
12
13 int MPI_Buffer_detach_c(void *buffer,
14                      MPI_Count *size)

```

MPI_Bsend is blocking, meaning that the routine will assume the message is fully contained in a buffer provided that the length is specified by the user through the function **MPI_Buffer_attach**.

MPI_Buffer_attach Links the MPI buffer to a predefined vector, with a fixed size. If **size** larger than a 32-bit integer use **MPI_Buffer_attach_c**.

MPI_Buffer_detach (**MPI_Buffer_detach_c**) returns the pointers to the buffer vector and buffer size memory.

```
1 int MPI_Ssend(const void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int destination_rank,  
5             int tag,  
6             MPI_Comm comm);
```

```
1 int MPI_Rsend(const void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int destination_rank,  
5             int tag,  
6             MPI_Comm comm);
```

MPI_Ssend has explicit blocking, similar to *standard* but it **will not return until the matching receive has been pointed**.

MPI_Rsend **crashes if the receiver is not READY TO RECEIVE** the communication.

All subroutines start with the capital 'I' that stands for **Instantaneous**

```
1 int MPI_I(,b,s,r)send(const void *data,
2                     int data_size,
3                     MPI_Datatype data_type,
4                     int destination_rank,
5                     int tag,
6                     MPI_Comm comm,
7                     MPI_Request *request);
8
```

```
1 // Wait until transfer done
2 int MPI_Wait(MPI_Request *request,
3             MPI_Status *status)
4
5 // Test transfer and proceed
6 int MPI_Test(MPI_Request *request,
7             int *flag,
8             MPI_Status *status)
```

All `MPI_I(,b,s,r)send` **return immediately, thus the code continues.**

`MPI_Wait` waits for the send to be executed correctly (according to the type of send).

`MPI_Test` Test the specific requests, `flag` returns the status.

The `status` can be simply set in both cases to `MPI_STATUS_IGNORE`.

Non-blocking Recv (MPI_Irecv)

MPI Send and MPI Recv Synchronizations

Mode	Function	Key Mechanism	Common Failures
Standard	MPI_Send	MPI chooses buffered	Deadlock if buffer is insufficient and receive is not executed.
Buffered	MPI_Bsend	Uses explicit user defined MPI buffer.	Buffer Overflow: Fails or crashes if attached buffer size is insufficient or missing MPI_BSEND_OVERHEAD.
Synchronous	MPI_Ssend	Completes only when matching MPI_Recv starts.	Deadlock if receiving rank never calls MPI_Recv or if order is mismatched.
Ready	MPI_Rsend	Assumes receiver is already waiting.	Crashes or silently drops data if matching MPI_Recv was not called beforehand.
Non-blocking	MPI_I*send	Initiates send asynchronously, returns MPI_Request.	Can modify data before MPI_Wait or MPI_Test completes (Data Race).

We can also define *non-blocking standard receive*

```
1 int MPI_Irecv(void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int source_rank,  
5             int tag,  
6             MPI_Comm comm,  
7             MPI_Request *request);
```

```
1 int MPI_Recv(void *data,  
2             int data_size,  
3             MPI_Datatype data_type,  
4             int source_rank,  
5             int tag,  
6             MPI_Comm comm,  
7             MPI_Status *status);
```

The function again **returns immediately**, thus the code continues, and the receive operation must be verified through `MPI_Wait` or `MPI_Test`.

The functions `MPI_recv` and `MPI_Irecv` also accept the special variables `MPI_ANY_SOURCE` for the `source_rank` argument, which allows a process to receive a message from any source, and the variable `MPI_ANY_TAG` for the tag argument, thus accepting all sources.

When using non-blocking or blocking communications, it is possible to verify the status of the send operation through the instructions `MPI_probe` or `MPI_Iprobe`.

```
1 int MPI_Probe(int source_rank,
2             int tag,
3             MPI_Comm comm,
4             MPI_Status *status);

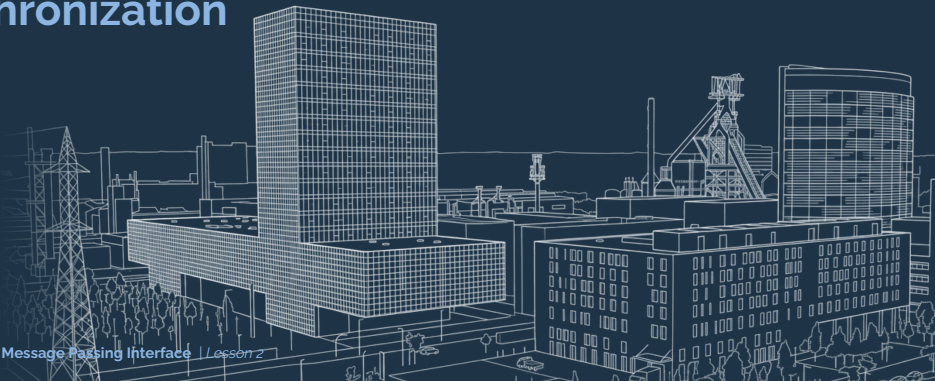
1 int MPI_Iprobe(int source_rank,
2               int tag,
3               MPI_Comm comm,
4               int *flag,
5               MPI_Status *status);
```

If in general the length of the data is unknown, it is possible to use the function `MPI_Get_count` to determine the number of elements received in a message.

```
1 int MPI_Get_count(const MPI_Status *status,
2                  MPI_Datatype data_type,
3                  int *count);
```

Here `count` is the number of received elements of the `data` communicated to the specific process and probed through `MPI_Probe` calls.

MPI Synchronization



Blocks the caller until all processes in the communicator have called it.

```
1 int MPI_Barrier(MPI_Comm comm);  
2
```

It is used to synchronize processes at a specific execution point.

It ensures that all processes reach the same point and thus the same data state before proceeding.

No process can cross the barrier until **all** processes arrive.

```
1 // All processes execute their operations with different timings  
2  
3 . . .  
4  
5 // Barrier Ensure all processes finish before measuring time  
6 MPI_Barrier(MPI_COMM_WORLD);  
7
```

Overusing barriers causes idle processor time (waiting for the slowest process) and degrades overall performance.

Collective communications (e.g., **MPI_Reduce**) already handle necessary synchronization internally.

Exercise: Simple Dot product of two vectors



The dot product between two vectors

Exercise: Simple Dot product of two vectors

We recall that the dot product between two double precision vectors **a** and **b**, both of length N is defined as:

$$c = \mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^N a_i b_i. \quad (1)$$

We define the simple serial code algorithm as

```
1  int num_of_elements=100;
2  double *a;
3  double *b;
4  double c = 0.0;
5  a = (double *)calloc(num_of_elements, sizeof(double));
6  b = (double *)calloc(num_of_elements, sizeof(double));
7  for (int i = 0; i < num_of_elements; i++)
8  {
9      a[i] = (double)i;
10     b[i] = (double)i;
11 }
12 for (int i = 0; i < num_of_elements; i++)
13 {
14     c+=a[i]*b[i];
15 }
16 printf(" %4.2f ", c);
```

Consider two vectors **a** and **b** of N elements each, compute their dot product defined as

$$c = \mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^N a_i b_i.$$

Parallelize the operation with MPI_Send and MPI_Recv.

1. Rank 0 divides the vectors in M segments equal to the total number of MPI processes. (Warning!! One rank might have more or less elements)
2. Rank 0 communicates the value of the lengths of each segment to the other ranks.
3. Rank 0 communicates the segments to each corresponding rank.
4. Each process multiplies the partial vectors. They do separate dot products.
5. Collect all values from each process to the main one, (rank 0)
6. Rank 0 sums the collected values and prints the integral.